

VELAIRE Phase 1 – Technologies & Software Used

Authentication & Communication Tools

- **Google & Apple OAuth** – Sign in with Google/Apple for secure and seamless authentication.
 - **Resend** – Email service provider (2,500 free emails/month).
 - **Twilio** – SMS messaging and 10DLC compliance for verified communication.
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Hosting & Deployment

- **Render** – Free backend hosting.
 - **Vercel** – Free frontend hosting with CI/CD support for React + Vite apps.
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AI/ML Infrastructure

- **FastAPI** – Lightweight and high-performance Python backend for ML APIs.
 - **LangChain** – Framework for building LLM-based agents and workflows.
 - **OpenAI** – For GPT-based reasoning, responses, and language understanding.
 - **Ollama** – Local model hosting and inference for LLMs.
 - **Hugging Face** – For pre-trained models and embeddings.
 - **ChromaDB** – Vector database for semantic search.
 - **MongoDB** – Storing parsed and structured document data.
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Frontend Stack

- **React.js + Vite** – Lightning-fast frontend with component-based architecture and modern tooling.
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Backend Stack

- **Node.js** – Event-driven backend service layer.
 - **MongoDB** – Primary NoSQL data storage.
 - **Redis** – For caching and fast in-memory operations.
 - **RabbitMQ** – Message queue for async task distribution.
 - **Socket.IO** – Real-time bidirectional communication (chat/messages).
 - **Multer** – File upload middleware (monitoring uploads).
 - **Cloudinary** – Secure and scalable file/image hosting and delivery.
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